Class 9: Candy Mini-Project

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Background

Today we are takina a detour to analyze a fun dataset (that we have more intrinsic insight into) with the most useful analysis method we have learned thus far - Principal Compenent Analysis (PCA)

The data is all related to halloween candy and is from the 538 website.

Importing candy data

```
candy_file <- "candy-data.csv"

candy = read.csv(candy_file, row.names=1)
head(candy)
```

```
             chocolate fruity caramel peanutyalmondy nougat crispedricewafer
100 Grand            1      0       1              0      0                1
3 Musketeers         1      0       0              0      1                0
One dime             0      0       0              0      0                0
One quarter          0      0       0              0      0                0
Air Heads            0      1       0              0      0                0
Almond Joy           1      0       0              1      0                0
             hard bar pluribus sugarpercent pricepercent winpercent
100 Grand       0   1        0        0.732        0.860   66.97173
3 Musketeers    0   1        0        0.604        0.511   67.60294
One dime        0   0        0        0.011        0.116   32.26109
One quarter     0   0        0        0.011        0.511   46.11650
Air Heads       0   0        0        0.906        0.511   52.34146
Almond Joy      0   1        0        0.465        0.767   50.34755
```

What is in the dataset?

> Q1. How many different candy types are in this dataset?

```r
nrow(candy)
```

```
[1] 85
```

=> there are 85 rows (which indicate different candy types) in this dataset.

> Q2. How many fruity candy types are in the dataset?

```r
nrow(candy[candy$fruity == 1, ])
```

```
[1] 38
```

```r
table(candy$fruity)
```

```

 0  1 
47 38 
```

=> There are 38 fruity candy types in the dataset.

What is your favorite candy?

```r
candy["Twix", ]$winpercent
```

```
[1] 81.64291
```

```r
library(dplyr)
```

```

Attaching package: 'dplyr'

The following objects are masked from 'package:stats':

    filter, lag

The following objects are masked from 'package:base':

    intersect, setdiff, setequal, union
```

```r
candy |>
  filter(row.names(candy)=="Twix") |>
  select(winpercent)
```

```
     winpercent
Twix   81.64291
```

> Q3. What is your favorite candy (other than Twix) in the dataset and what is it's winpercent value?

```r
candy["Junior Mints", ]$winpercent
```

```
[1] 57.21925
```

=> 57.21925

> Q4. What is the winpercent value for "Kit Kat"?

```
candy["Kit Kat", ]$winpercent
```

```
[1] 76.7686
```

=> 76.7686

Q5. What is the winpercent value for "Tootsie Roll Snack Bars"?

```
candy["Tootsie Roll Snack Bars", ]$winpercent
```

```
[1] 49.6535
```

=> 49.6535

Side-note: the skimr::skim() function

```
#library("skimr")
skimr::skim(candy)
```

Table 1: Data summary

Name	candy
Number of rows	85
Number of columns	12
Column type frequency:	
numeric	12
Group variables	None

Variable type: numeric

skim_variable	n_missing	complete_rate	mean	sd	p0	p25	p50	p75	p100	hist
chocolate	0	1	0.44	0.50	0.00	0.00	0.00	1.00	1.00	
fruity	0	1	0.45	0.50	0.00	0.00	0.00	1.00	1.00	
caramel	0	1	0.16	0.37	0.00	0.00	0.00	0.00	1.00	

skim_variable	n_missing	complete_rate	mean	sd	p0	p25	p50	p75	p100	hist
peanutyalmondy	0	1	0.16	0.37	0.00	0.00	0.00	0.00	1.00	
nougat	0	1	0.08	0.28	0.00	0.00	0.00	0.00	1.00	
crispedricewafer	0	1	0.08	0.28	0.00	0.00	0.00	0.00	1.00	
hard	0	1	0.18	0.38	0.00	0.00	0.00	0.00	1.00	
bar	0	1	0.25	0.43	0.00	0.00	0.00	0.00	1.00	
pluribus	0	1	0.52	0.50	0.00	0.00	1.00	1.00	1.00	
sugarpercent	0	1	0.48	0.28	0.01	0.22	0.47	0.73	0.99	
pricepercent	0	1	0.47	0.29	0.01	0.26	0.47	0.65	0.98	
winpercent	0	1	50.32	14.71	22.45	39.14	47.83	59.86	84.18	

> Q6. Is there any variable/column that looks to be on a different scale to the majority of the other columns in the dataset?

9 variables, from `chocolate` to `pluribus`, has a value of either 0 or 1. (`int` type) Other 3 variables that end with `percent` is a `numeric` type value. These store an actual number rather than categorical assignment as other 9 variables.

And among those 3 variables, `winpercent` has a different scale because values for other "percent" variables are between 0 and 1, while this variable has values between 0 and 100.

> Q7. What do you think a zero and one represent for the candy$chocolate column?

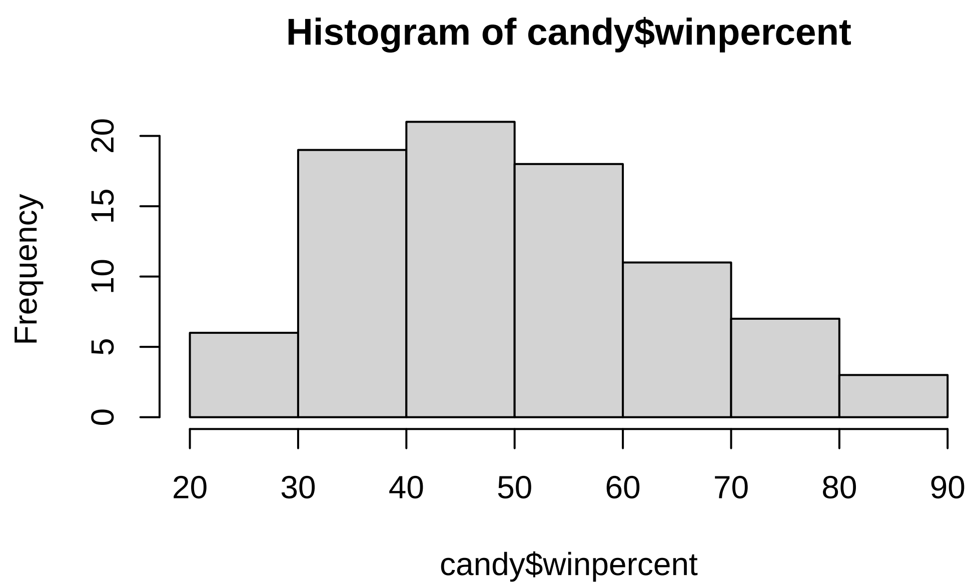
I think it is basically same as a logical value -> `0` represents `False` and `1` represents `True`. So for the candy$chocolate column, `0` would indicate that the candy type is **not a chocolate (type candy)** and `1` would indicate that it **is a chocolate (type candy)**

Exploratory analysis

> Q8. Plot a histogram of winpercent values using both base R and ggplot2.

Using base R:

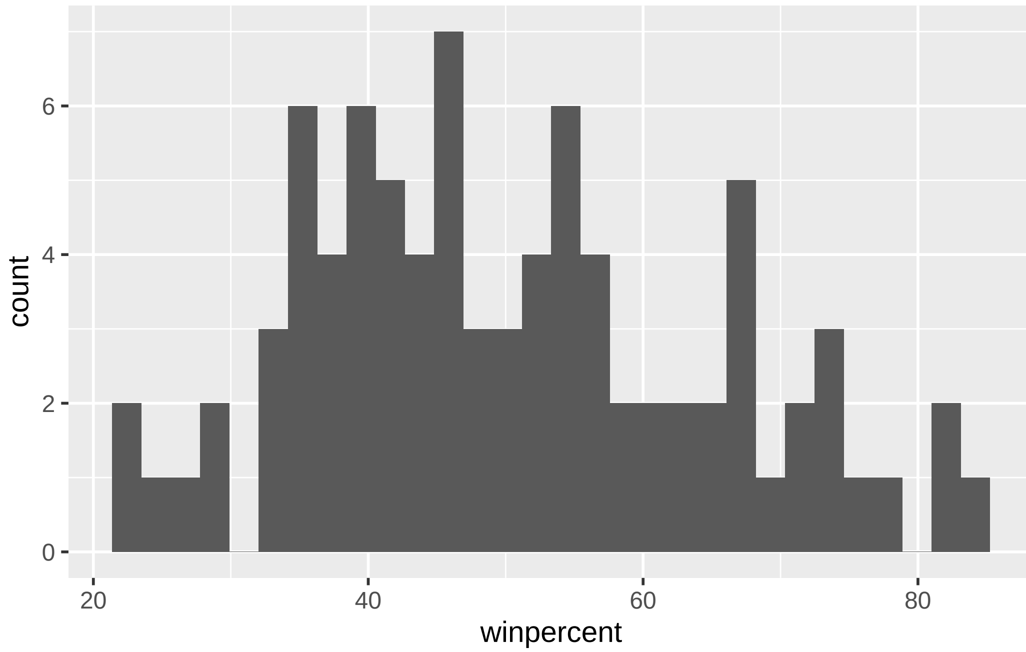
```
hist(candy$winpercent)
```



Using ggplot2:

```
library(ggplot2)
ggplot (candy, aes(winpercent)) +
  geom_histogram()
```

```
`stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```



Q9. Is the distribution of winpercent values symmetrical?

From the plots in Q8, **No**.

Q10. Is the center of the distribution above or below 50%?

Center of the distribution would be:

```
median(candy$winpercent)
```

```
[1] 47.82975
```

which is **below** 50%.

Q11. On average is chocolate candy higher or lower ranked than fruit candy?

```
chocolate_winpercent <- mean(candy$winpercent[candy$chocolate == 1])
fruity_winpercent <- mean(candy$winpercent[candy$fruity == 1])
cat ("chocolate_winpercent", chocolate_winpercent, "\n")
```

```
chocolate_winpercent 60.92153
```

```r
cat ("fruity_winpercent", fruity_winpercent, "\n")
```

```
fruity_winpercent 44.11974
```

```r
chocolate_winpercent > fruity_winpercent
```

```
[1] TRUE
```

=> Chocolate candies have **higher** winpercent average than fruity candies.

Using `as.logical()`:

```r
mean(candy$winpercent[as.logical(candy$chocolate)])
```

```
[1] 60.92153
```

```r
mean(candy$winpercent[as.logical(candy$fruity)])
```

```
[1] 44.11974
```

> Q12. Is this difference statistically significant?

```r
choc_winperc_data <- candy$winpercent[as.logical(candy$chocolate)]
fruit_winperc_data <- candy$winpercent[as.logical(candy$fruity)]

t.test (choc_winperc_data, fruit_winperc_data)
```

```

	Welch Two Sample t-test

data:  choc_winperc_data and fruit_winperc_data
t = 6.2582, df = 68.882, p-value = 2.871e-08
alternative hypothesis: true difference in means is not equal to 0
95 percent confidence interval:
 11.44563 22.15795
sample estimates:
mean of x mean of y 
 60.92153  44.11974 
```

=> The difference is statistically significant, as it has a very low p-value («0.05). This indicates that the null hypothesis is rejected and that the observed difference is likely not due to chance.

Overall Candy Rankings

> Q13. What are the five least liked candy types in this set?

Using base R:

```
head(candy[order(candy$winpercent),], n=5)
```

```
                   chocolate fruity caramel peanutyalmondy nougat
Nik L Nip                  0      1       0              0      0
Boston Baked Beans         0      0       0              1      0
Chiclets                   0      1       0              0      0
Super Bubble               0      1       0              0      0
Jawbusters                 0      1       0              0      0
                   crispedricewafer hard bar pluribus sugarpercent pricepercent
Nik L Nip                         0    0   0        1        0.197        0.976
Boston Baked Beans                0    0   0        1        0.313        0.511
Chiclets                          0    0   0        1        0.046        0.325
Super Bubble                      0    0   0        0        0.162        0.116
Jawbusters                        0    1   0        1        0.093        0.511
                   winpercent
Nik L Nip            22.44534
Boston Baked Beans   23.41782
Chiclets             24.52499
Super Bubble         27.30386
Jawbusters           28.12744
```

Using dplyr:

```
candy |>
  arrange(winpercent) |>
  head(5)
```

```
                   chocolate fruity caramel peanutyalmondy nougat
Nik L Nip                  0      1       0              0      0
Boston Baked Beans         0      0       0              1      0
Chiclets                   0      1       0              0      0
Super Bubble               0      1       0              0      0
Jawbusters                 0      1       0              0      0
                   crispedricewafer hard bar pluribus sugarpercent pricepercent
Nik L Nip                         0    0   0        1        0.197        0.976
```

```
Boston Baked Beans                0     0    0          1         0.313          0.511
Chiclets                          0     0    0          1         0.046          0.325
Super Bubble                      0     0    0          0         0.162          0.116
Jawbusters                        0     1    0          1         0.093          0.511
                   winpercent
Nik L Nip            22.44534
Boston Baked Beans   23.41782
Chiclets             24.52499
Super Bubble         27.30386
Jawbusters           28.12744
```

=> **Nik L Nip, Boston Baked Beans, Chiclets, Super Bubble, Jawbusters**

Q14. What are the top 5 all time favorite candy types out of this set?

```
head(candy[order(candy$winpercent, decreasing=TRUE),], n=5)
```

```
                          chocolate fruity caramel peanutyalmondy nougat
Reese's Peanut Butter cup         1      0       0              1      0
Reese's Miniatures                1      0       0              1      0
Twix                              1      0       1              0      0
Kit Kat                           1      0       0              0      0
Snickers                          1      0       1              1      1
                          crispedricewafer hard bar pluribus sugarpercent
Reese's Peanut Butter cup                0    0   0        0        0.720
Reese's Miniatures                       0    0   0        0        0.034
Twix                                     1    0   1        0        0.546
Kit Kat                                  1    0   1        0        0.313
Snickers                                 0    0   1        0        0.546
                          pricepercent winpercent
Reese's Peanut Butter cup        0.651   84.18029
Reese's Miniatures               0.279   81.86626
Twix                             0.906   81.64291
Kit Kat                          0.511   76.76860
Snickers                         0.651   76.67378
```

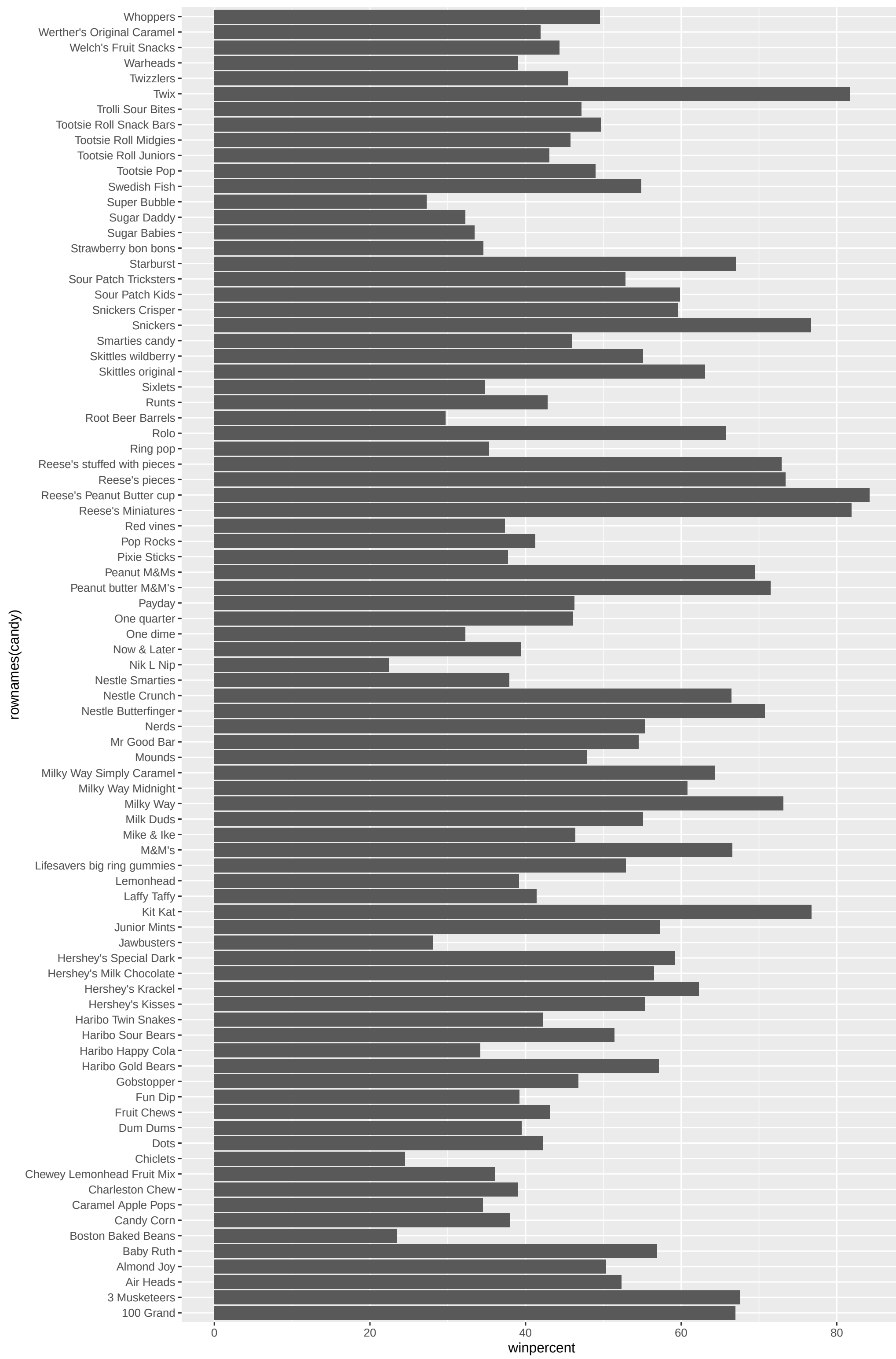
```
candy |>
  arrange(desc(winpercent)) |>
  head(5)
```

```
                           chocolate fruity caramel peanutyalmondy nougat
Reese's Peanut Butter cup          1      0       0              1      0
Reese's Miniatures                 1      0       0              1      0
Twix                               1      0       1              0      0
Kit Kat                            1      0       0              0      0
Snickers                           1      0       1              1      1
                           crispedricewafer hard bar pluribus sugarpercent
Reese's Peanut Butter cup                 0    0   0        0        0.720
Reese's Miniatures                        0    0   0        0        0.034
Twix                                      1    0   1        0        0.546
Kit Kat                                   1    0   1        0        0.313
Snickers                                  0    0   1        0        0.546
                           pricepercent winpercent
Reese's Peanut Butter cup         0.651   84.18029
Reese's Miniatures                0.279   81.86626
Twix                              0.906   81.64291
Kit Kat                           0.511   76.76860
Snickers                          0.651   76.67378
```

=> **Reese's Peanut Butter cup, Reese's Miniatures, Twix, Kit Kat, Snickers**

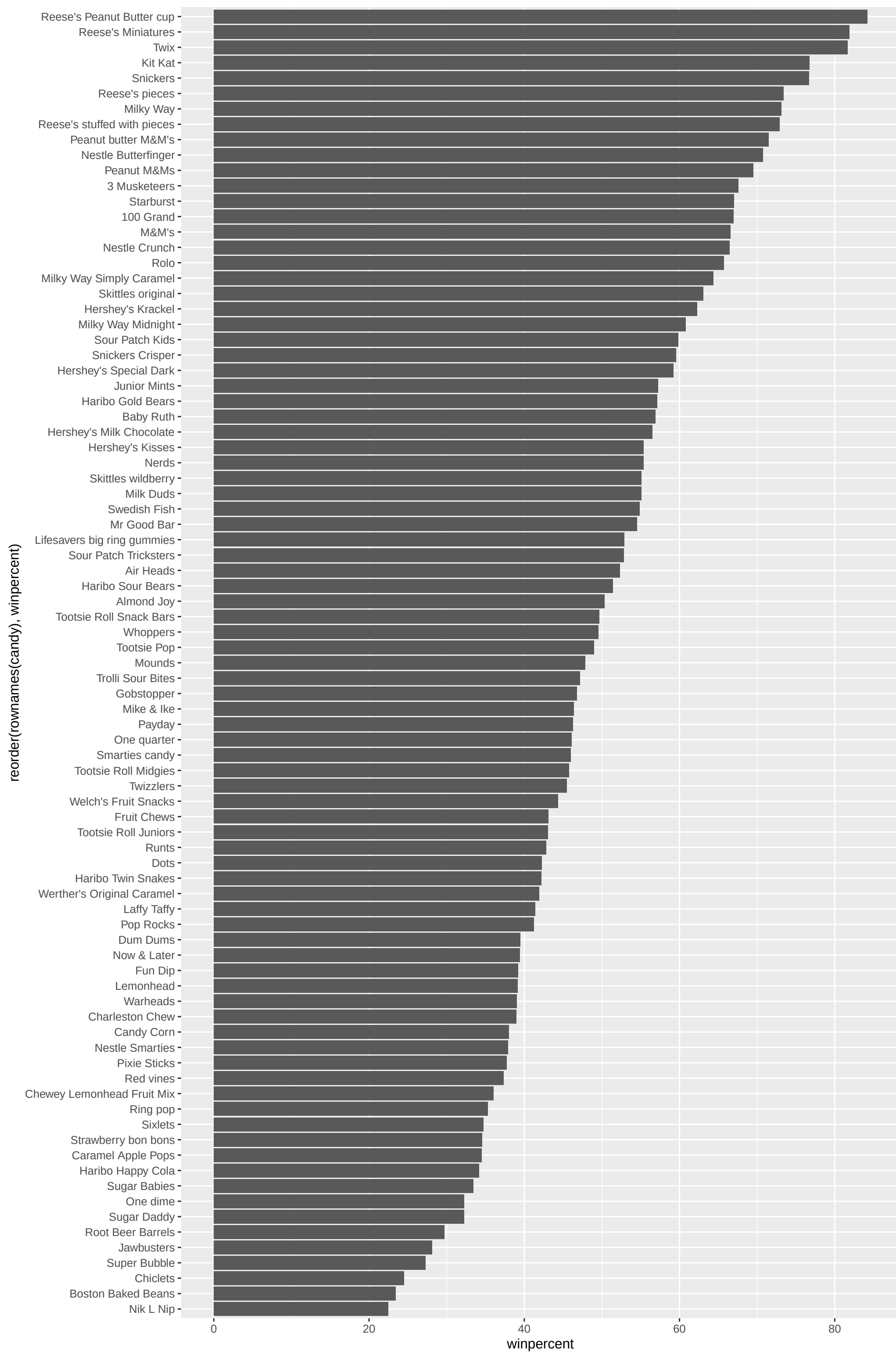
Q15. Make a first barplot of candy ranking based on winpercent values.

```r
library(ggplot2)

ggplot(candy) +
  aes(winpercent, rownames(candy)) +
  geom_col()
```


Whoppers
Werther's Original Caramel
Welch's Fruit Snacks
Warheads
Twizzlers
Twix
Trolli Sour Bites
Tootsie Roll Snack Bars
Tootsie Roll Midgies
Tootsie Roll Juniors
Tootsie Pop
Swedish Fish
Super Bubble
Sugar Daddy
Sugar Babies
Strawberry bon bons
Starburst
Sour Patch Tricksters
Sour Patch Kids
Snickers Crisper
Snickers
Smarties candy
Skittles wildberry
Skittles original
Sixlets
Runts
Root Beer Barrels
Rolo
Ring pop
Reese's stuffed with pieces
Reese's pieces
Reese's Peanut Butter cup
Reese's Miniatures
Red vines
Pop Rocks
Pixie Sticks
Peanut M&Ms
Peanut butter M&M's
Payday
One quarter
One dime
Now & Later
Nik L Nip
Nestle Smarties
Nestle Crunch
Nestle Butterfinger
Nerds
Mr Good Bar
Mounds
Milky Way Simply Caramel
Milky Way Midnight
Milky Way
Milk Duds
Mike & Ike
M&M's
Lifesavers big ring gummies
Lemonhead
Laffy Taffy
Kit Kat
Junior Mints
Jawbusters
Hershey's Special Dark
Hershey's Milk Chocolate
Hershey's Krackel
Hershey's Kisses
Haribo Twin Snakes
Haribo Sour Bears
Haribo Happy Cola
Haribo Gold Bears
Gobstopper
Fun Dip
Fruit Chews
Dum Dums
Dots
Chiclets
Chewey Lemonhead Fruit Mix
Charleston Chew
Caramel Apple Pops
Candy Corn
Boston Baked Beans
Baby Ruth
Almond Joy
Air Heads
3 Musketeers
100 Grand
rownames(candy)
0
20
40
60
80
winpercent

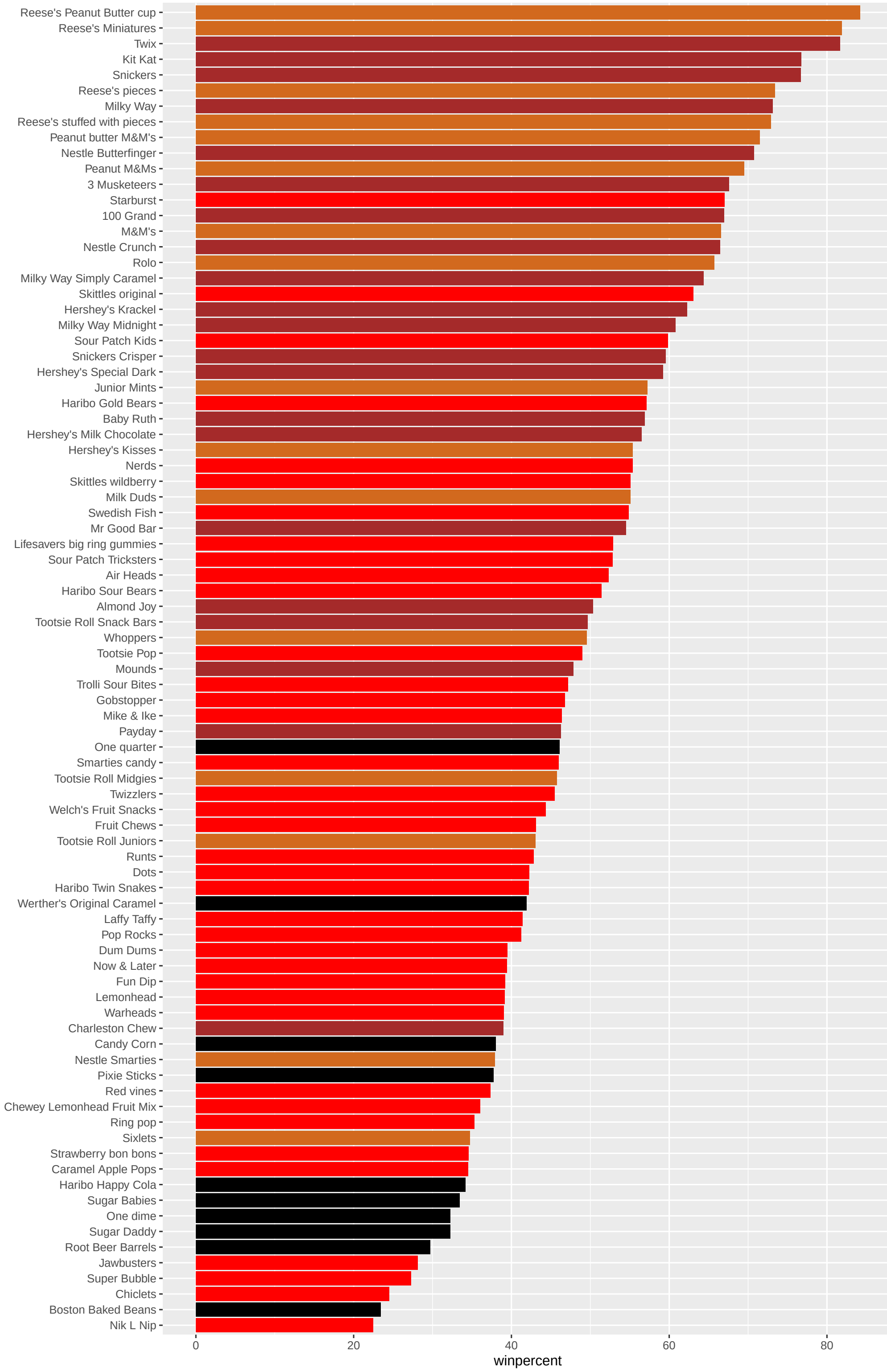
Q16. This is quite ugly, use the reorder() function to get the bars sorted by winpercent?

```
library(ggplot2)

ggplot(candy) +
  aes(winpercent, reorder(rownames(candy),winpercent)) +
  geom_col()
```


reorder(rownames(candy), winpercent)
Reese's Peanut Butter cup
Reese's Miniatures
Twix
Kit Kat
Snickers
Reese's pieces
Milky Way
Reese's stuffed with pieces
Peanut butter M&M's
Nestle Butterfinger
Peanut M&Ms
3 Musketeers
Starburst
100 Grand
M&M's
Nestle Crunch
Rolo
Milky Way Simply Caramel
Skittles original
Hershey's Krackel
Milky Way Midnight
Sour Patch Kids
Snickers Crisper
Hershey's Special Dark
Junior Mints
Haribo Gold Bears
Baby Ruth
Hershey's Milk Chocolate
Hershey's Kisses
Nerds
Skittles wildberry
Milk Duds
Swedish Fish
Mr Good Bar
Lifesavers big ring gummies
Sour Patch Tricksters
Air Heads
Haribo Sour Bears
Almond Joy
Tootsie Roll Snack Bars
Whoppers
Tootsie Pop
Mounds
Trolli Sour Bites
Gobstopper
Mike & Ike
Payday
One quarter
Smarties candy
Tootsie Roll Midgies
Twizzlers
Welch's Fruit Snacks
Fruit Chews
Tootsie Roll Juniors
Runts
Dots
Haribo Twin Snakes
Werther's Original Caramel
Laffy Taffy
Pop Rocks
Dum Dums
Now & Later
Fun Dip
Lemonhead
Warheads
Charleston Chew
Candy Corn
Nestle Smarties
Pixie Sticks
Red vines
Chewey Lemonhead Fruit Mix
Ring pop
Sixlets
Strawberry bon bons
Caramel Apple Pops
Haribo Happy Cola
Sugar Babies
One dime
Sugar Daddy
Root Beer Barrels
Jawbusters
Super Bubble
Chiclets
Boston Baked Beans
Nik L Nip
0
20
40
60
80
winpercent

Time to add some useful color

```
my_cols=rep("black", nrow(candy))
my_cols[as.logical(candy$chocolate)] = "chocolate"
my_cols[as.logical(candy$bar)] = "brown"
my_cols[as.logical(candy$fruity)] = "red"

ggplot(candy) +
  aes(winpercent, reorder(rownames(candy),winpercent)) +
  geom_col(fill=my_cols) +
  ylab("")
```


Reese's Peanut Butter cup
Reese's Miniatures
Twix
Kit Kat
Snickers
Reese's pieces
Milky Way
Reese's stuffed with pieces
Peanut butter M&M's
Nestle Butterfinger
Peanut M&Ms
3 Musketeers
Starburst
100 Grand
M&M's
Nestle Crunch
Rolo
Milky Way Simply Caramel
Skittles original
Hershey's Krackel
Milky Way Midnight
Sour Patch Kids
Snickers Crisper
Hershey's Special Dark
Junior Mints
Haribo Gold Bears
Baby Ruth
Hershey's Milk Chocolate
Hershey's Kisses
Nerds
Skittles wildberry
Milk Duds
Swedish Fish
Mr Good Bar
Lifesavers big ring gummies
Sour Patch Tricksters
Air Heads
Haribo Sour Bears
Almond Joy
Tootsie Roll Snack Bars
Whoppers
Tootsie Pop
Mounds
Trolli Sour Bites
Gobstopper
Mike & Ike
Payday
One quarter
Smarties candy
Tootsie Roll Midgies
Twizzlers
Welch's Fruit Snacks
Fruit Chews
Tootsie Roll Juniors
Runts
Dots
Haribo Twin Snakes
Werther's Original Caramel
Laffy Taffy
Pop Rocks
Dum Dums
Now & Later
Fun Dip
Lemonhead
Warheads
Charleston Chew
Candy Corn
Nestle Smarties
Pixie Sticks
Red vines
Chewey Lemonhead Fruit Mix
Ring pop
Sixlets
Strawberry bon bons
Caramel Apple Pops
Haribo Happy Cola
Sugar Babies
One dime
Sugar Daddy
Root Beer Barrels
Jawbusters
Super Bubble
Chiclets
Boston Baked Beans
Nik L Nip
0
20
40
60
80
winpercent

Q17. What is the worst ranked chocolate candy?

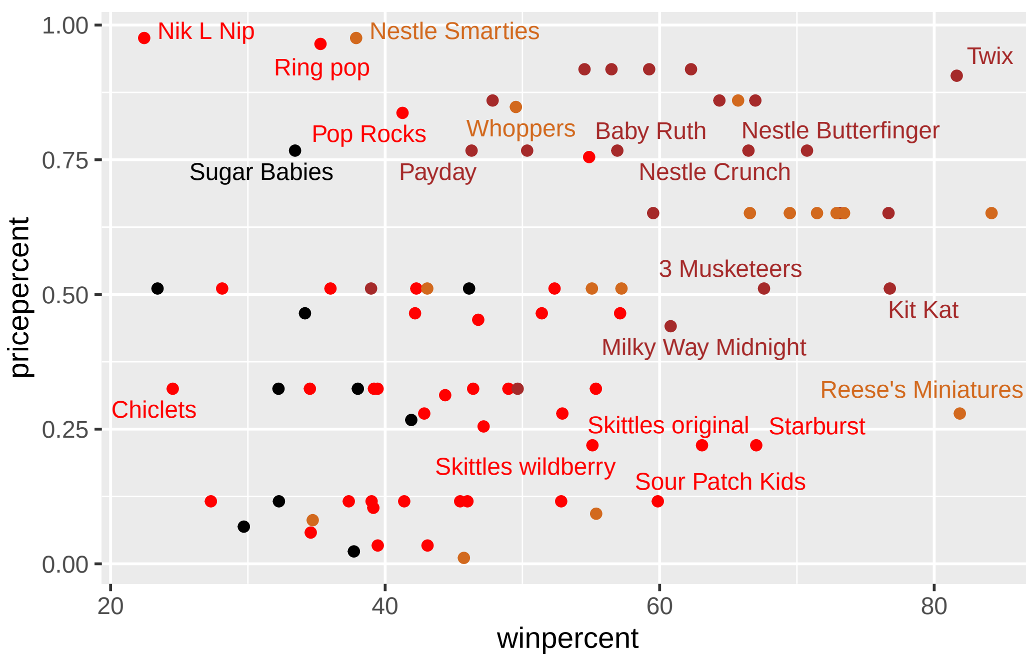
=> **Sixlets**

Q18. What is the best ranked fruity candy?

=> **Starburst**

Taking a look at pricepercent

```
library(ggrepel)

# How about a plot of win vs price
ggplot(candy) +
  aes(winpercent, pricepercent, label=rownames(candy)) +
  geom_point(col=my_cols) +
  geom_text_repel(col=my_cols, size=3.3, max.overlaps = 5)
```



Q19. Which candy type is the highest ranked in terms of winpercent for the least money - i.e. offers the most bang for your buck?

```r
ord <- order(candy$winpercent/candy$pricepercent, decreasing = TRUE)
head( candy[ord,c(11,12)], n=5 )
```

```
                     pricepercent winpercent
Tootsie Roll Midgies        0.011   45.73675
Pixie Sticks                0.023   37.72234
Fruit Chews                 0.034   43.08892
Dum Dums                    0.034   39.46056
Strawberry bon bons         0.058   34.57899
```

=> I sorted the candy types by `winpercent/pricepercent` value. The candy type with highest value is **Tootsie Roll Midgies**.

> Q20. What are the top 5 most expensive candy types in the dataset and of these which is the least popular?

Finding the top 5 most expensive candy types

Using base R:

```r
ord <- order(candy$pricepercent, decreasing = TRUE)
head( candy[ord,c(11,12)], n=5 )
```

```
                         pricepercent winpercent
Nik L Nip                       0.976   22.44534
Nestle Smarties                 0.976   37.88719
Ring pop                        0.965   35.29076
Hershey's Krackel               0.918   62.28448
Hershey's Milk Chocolate        0.918   56.49050
```

Using dplyr:

```r
library(dplyr)
candy |>
  arrange(-pricepercent) |>
  select(pricepercent, winpercent) |>
  head(n=5)
```

```
                          pricepercent winpercent
Nik L Nip                        0.976   22.44534
Nestle Smarties                  0.976   37.88719
Ring pop                         0.965   35.29076
Hershey's Krackel                0.918   62.28448
Hershey's Milk Chocolate         0.918   56.49050
```

=> the top 5 most expensive candy types are: **Nik L Nip, Nestle Smarties, Ring pop, Hershey's Krackel, Hershey's Milk Chocolate**.

=> from the table, we can say that **Nik L Nip** is the least popular among these five candy types.

Exploring the correlation structure

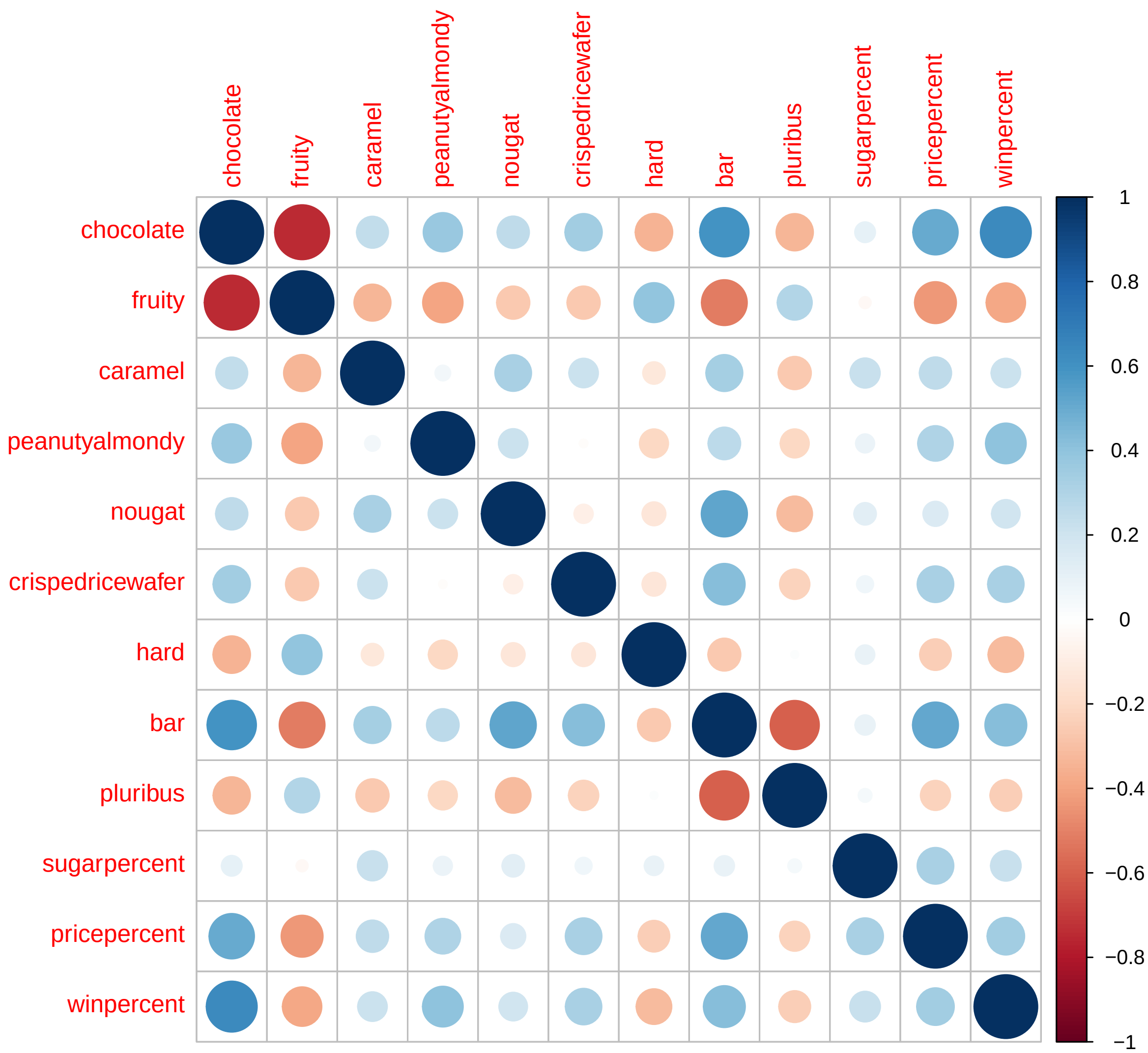
> From the sidenote: This is also why we will use scale=TRUE when calling prcomp() in the next section: it ensures all variables contribute equally by working from correlations (standardized and scale-free) rather than raw covariances, which would be dominated by whichever variable happens to have the largest numeric range. Can you guess which variable in our dataset that would be?

The variable that would dominate the `prcomp()` result without scale=TRUE argument setting would be the `winpercent` variable, as it has the largest numeric range in the dataset.

```
library(corrplot)
```

```
corrplot 0.95 loaded
```

```
cij <- cor(candy)
corrplot(cij)
```



Q22. Examining this plot what two variables are anti-correlated (i.e. have minus values)?

=> Some of the most evident anti-correlated pairs from this figure is: **chocolate-fruity, bar-fruity, bar-pluribus** and **fruity-pricepercent, fruity-winpercent, fruity-peanutyalmondy**, etc.

Q23. Use your corrplot result to make a prediction: which variables do you expect will have the largest contributions (i.e. loadings) to PC1 (i.e., drive the most separation between candies along the first principal component)?

fruity and chocolate. They have the largest magnitude of negative correlation, which shows that they would be likely to be the most "separated" variables in the dataset.

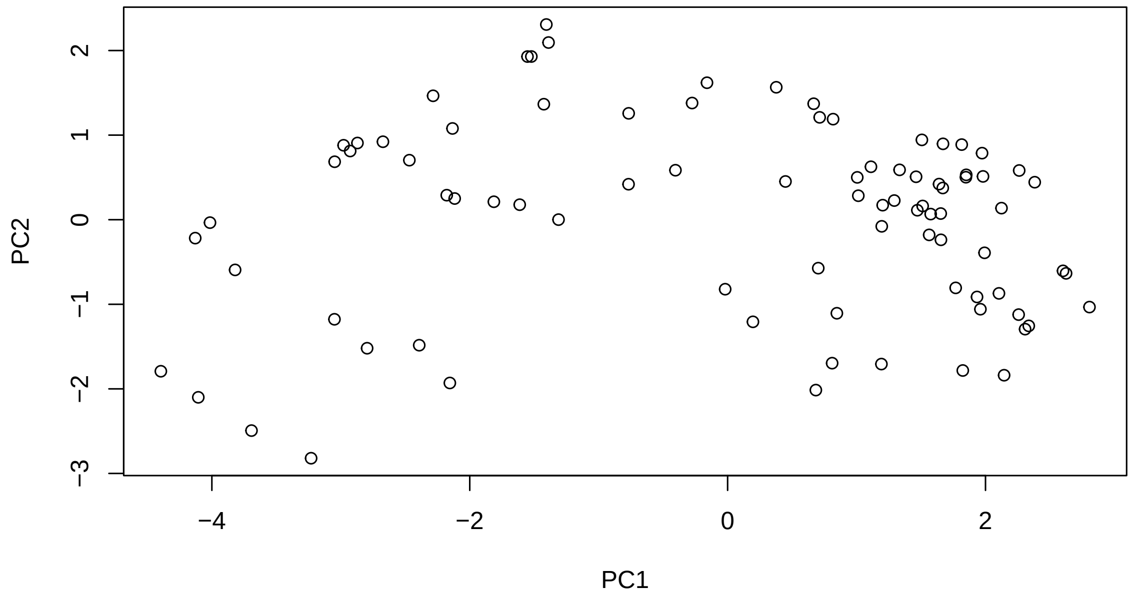
Principal Component Analysis

In this case we want to be sure to set `scale=TRUE` argument for `prcomp()` because we have one variable `winpercent`, that is on a very different scale than all others and would otherwise dominate our PCA results.

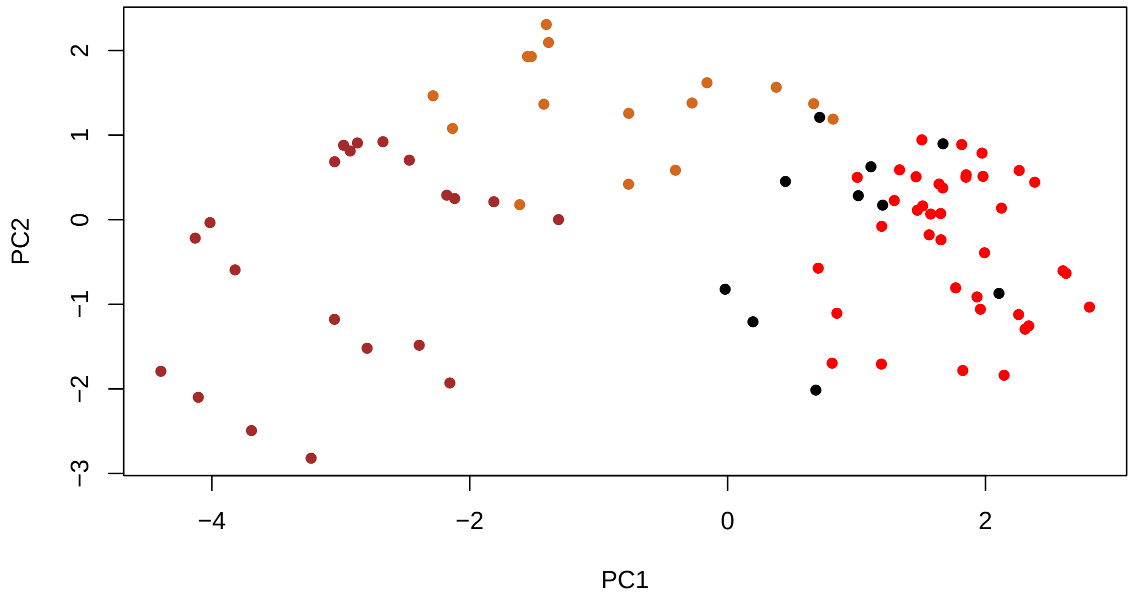
```
pca <- prcomp(candy, scale=TRUE)
summary(pca)
```

```
Importance of components:
                          PC1    PC2    PC3     PC4    PC5     PC6     PC7
Standard deviation     2.0788 1.1378 1.1092 1.07533 0.9518 0.81923 0.81530
Proportion of Variance 0.3601 0.1079 0.1025 0.09636 0.0755 0.05593 0.05539
Cumulative Proportion  0.3601 0.4680 0.5705 0.66688 0.7424 0.79830 0.85369
                           PC8     PC9    PC10    PC11    PC12
Standard deviation     0.74530 0.67824 0.62349 0.43974 0.39760
Proportion of Variance 0.04629 0.03833 0.03239 0.01611 0.01317
Cumulative Proportion  0.89998 0.93832 0.97071 0.98683 1.00000
```

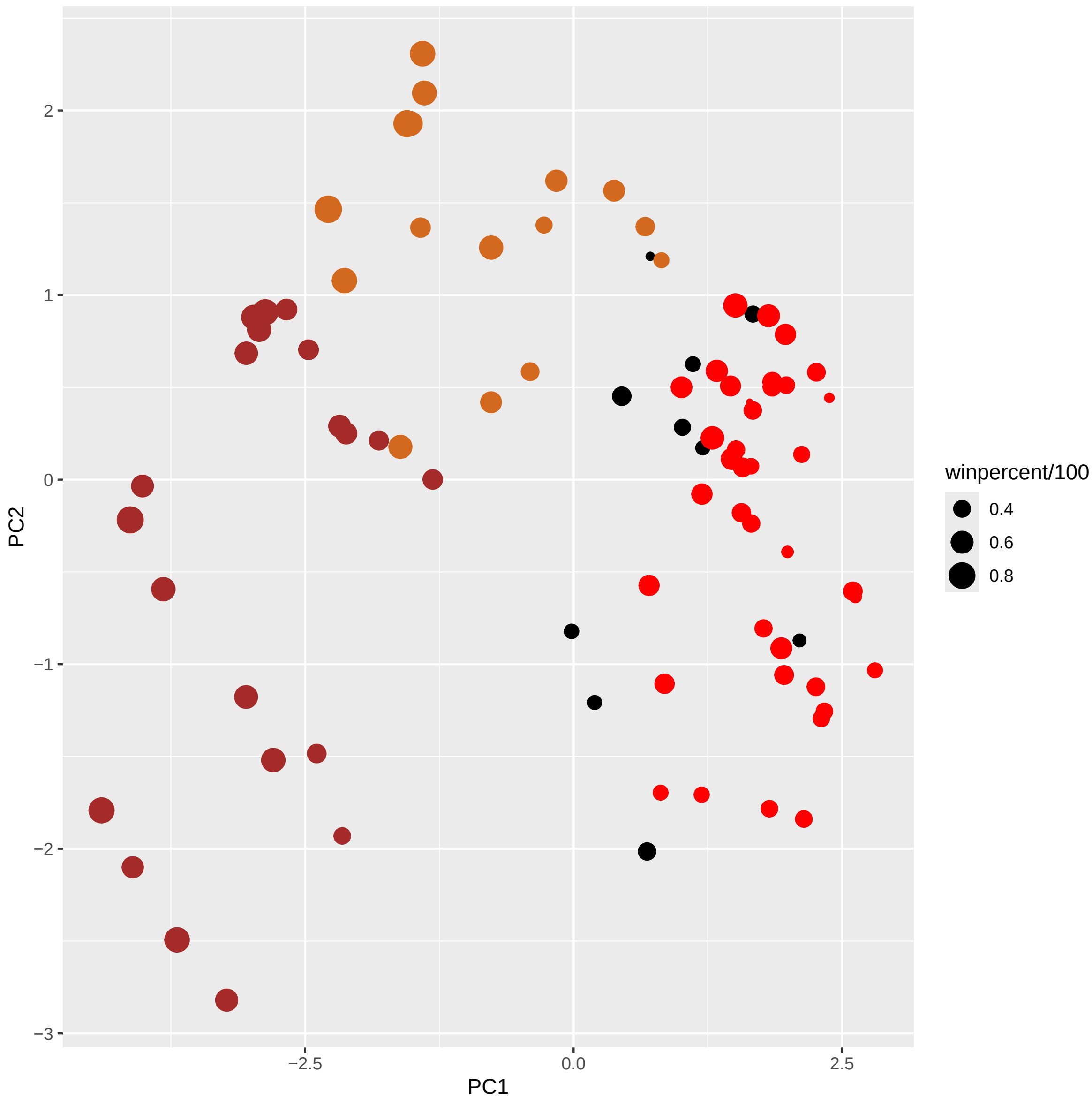
```
plot(pca$x[, c("PC1", "PC2")])
```



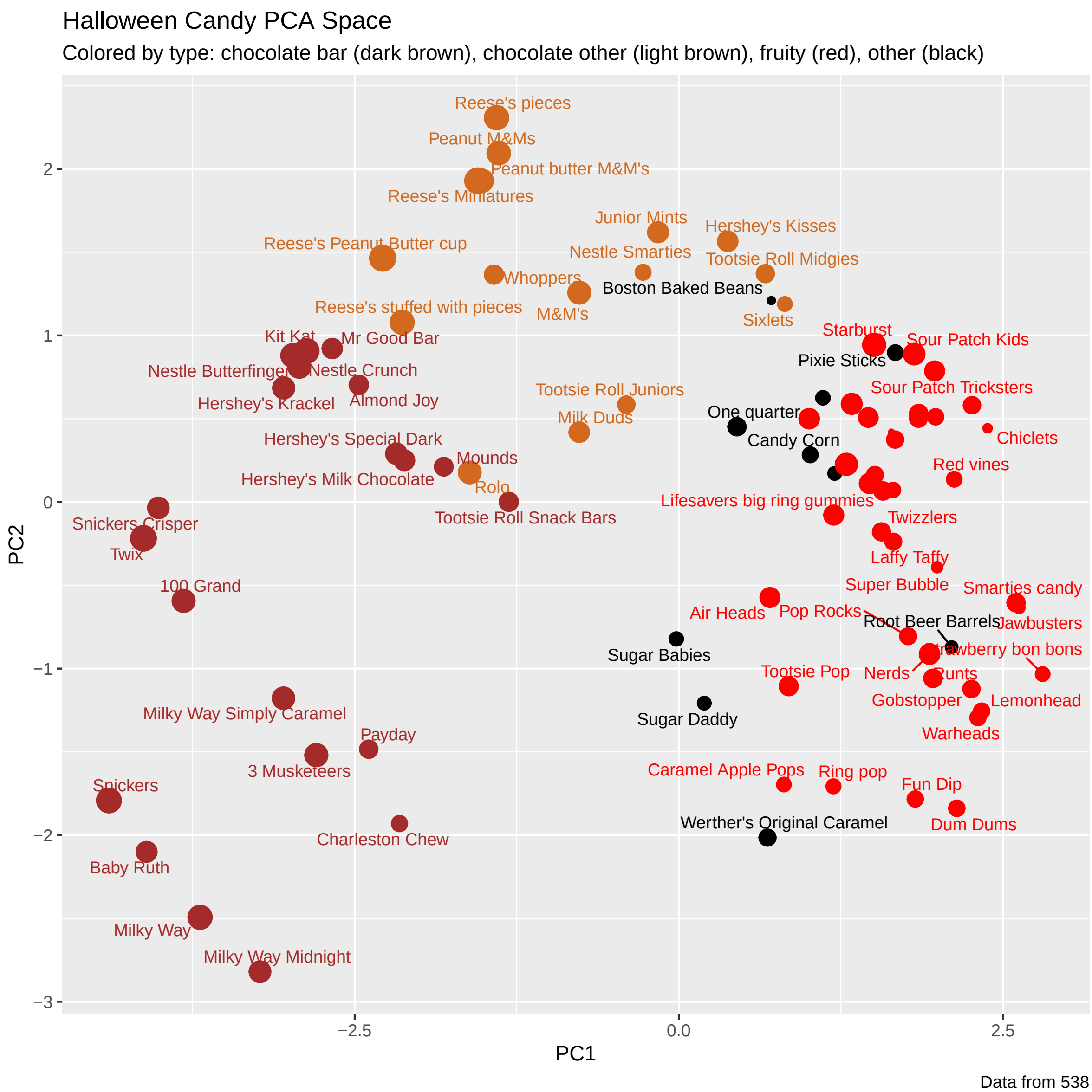

```
plot(pca$x[,1:2], col=my_cols, pch=16)
```




```
# Make a new data-frame with our PCA results and candy data
my_data <- cbind(candy, pca$x[,1:3])

p <- ggplot(my_data) +
        aes(x=PC1, y=PC2,
            size=winpercent/100,
            text=rownames(my_data),
            label=rownames(my_data)) +
        geom_point(col=my_cols)

p
```




```
library(ggrepel)

p + geom_text_repel(size=3.3, col=my_cols, max.overlaps = 7)  +
  theme(legend.position = "none") +
  labs(title="Halloween Candy PCA Space",
       subtitle="Colored by type: chocolate bar (dark brown), chocolate other (light brown),
       caption="Data from 538")
```

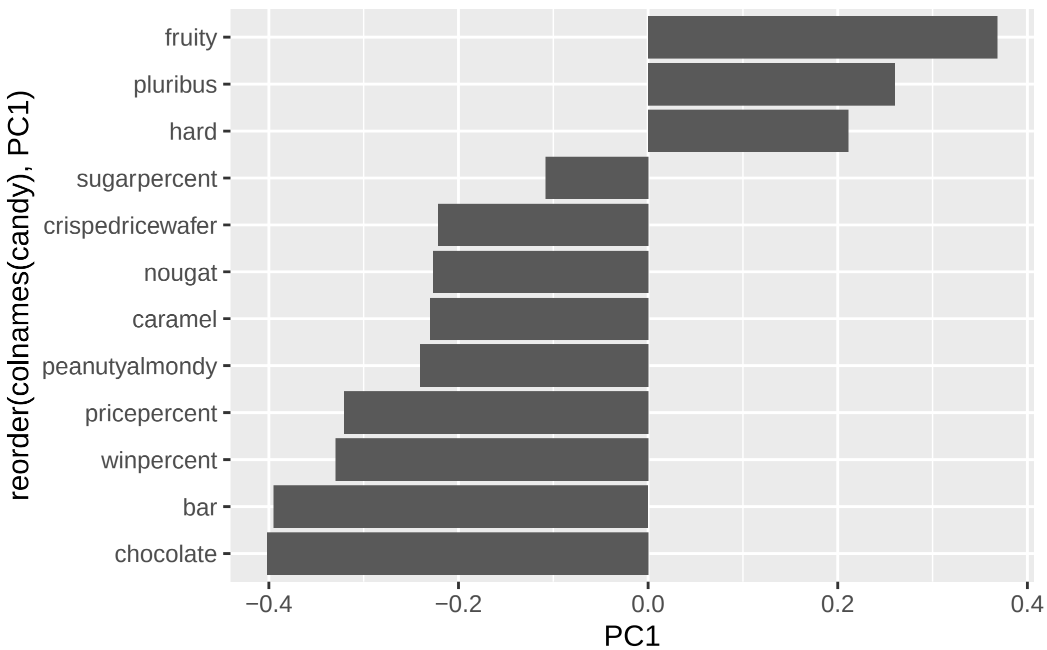


Using plotly to generate interactive plots

```
#library(plotly)
#ggplotly(p)
```

Q24. Complete the code to generate the loadings plot above. What original variables are picked up strongly by PC1 in the positive direction? Do these make sense to you? Where did you see this relationship highlighted previously?

```
ggplot(pca$rotation) +
  aes(PC1, reorder(colnames(candy),PC1)) +
  geom_col()
```



=> **fruity** has the most positive PC1 value among the variables. Other variables with positive PC1 value include `pluribus` and `hard`. Comparing this result with the previously generated **corrplot** *makes sense*, since `fruity` has shown positive correlation with `pluribus` and `hard`, while showing significant anti-correlation with the top negative variables in terms of PC1 - `chocolate`, `bar`, `winpercent`, `pricepercent` or `peanutyalmondy`. So the PCA captures large proportion of these complex relationships between variables in this axis PC1.

Summary

> Q25. Based on your exploratory analysis, correlation findings, and PCA results, what combination of characteristics appears to make a "winning" candy? How do these different analyses (visualization, correlation, PCA) support or complement each other in reaching this conclusion?

Making a "winning" candy - an **expensive peanutyalmondy chocolate bar**. characteristics fruity, hard, or pluribus should preferrably not be included here. Other attributes - caramel, nougat, crispedricewafer or sugarpercent - would have a positive effect (positive correlation) in making a winning candy, but the effects are much more moderate than the top 4 attributes indicated as bold.